BEGIN:

```
lab_16=# BEGIN;
BEGIN
lab_16=*# COMMIT;
COMMIT
lab_16=# BEGIN TRANSACTION;
BEGIN
lab_16=*# COMMIT;
COMMIT
lab_16=# START TRANSACTION;
START TRANSACTION
lab_16=*# COMMIT;
COMMIT
```

COMMIT:

```
lab_16=# BEGIN;
BEGIN
lab_16=*# UPDATE accounts SET balance = balance - 100 WHERE account_id = 1;
UPDATE 1
lab_16=*# UPDATE accounts SET balance = balance + 100 WHERE account_id = 2;
UPDATE 1
lab_16=*# COMMIT;
COMMIT
```

ROLLBACK:

```
lab_16=# BEGIN;
BEGIN
lab_16=*# UPDATE accounts SET balance = balance - 100 WHERE account_id = 1;
UPDATE 1
lab_16=*# UPDATE accounts SET balance = balance + 100 WHERE account_id = 2;
UPDATE 1
lab_16=*# ROLLBACK;
```

ATOMICITY:

```
lab_16=# BEGIN;
BEGIN
lab_16=*# INSERT INTO orders (customer_id, total) VALUES (1, 250.00);
INSERT 0 1
lab_16=*# INSERT INTO order_items (order_id, product_id, quantity) VALUES (1, 101, 2);
INSERT 0 1
lab_16=*# UPDATE inventory SET stock = stock - 2 WHERE product_id = 101;
UPDATE 1
lab_16=*# COMMIT;
COMMIT
lab_16=#
```

CONSISTENCY:

```
lab_16=# BEGIN;
BEGIN
lab_16=*# INSERT INTO customers (name, email) VALUES ('John Doe', 'john@email.com');
INSERT 0 1
lab_16=*# INSERT INTO orders (customer_id, total) VALUES (LASTVAL(), 100.00);
INSERT 0 1
lab_16=*# COMMIT;
COMMIT
```

READ UNCOMMITTED:

```
lab_16=# BEGIN TRANSACTION ISOLATION LEVEL READ UNCOMMITTED;
BEGIN
lab_16=*# SELECT * FROM accounts;
 account_id |      owner       | balance
------------+------------------+----------
          3 | Nurlan Bekov     |  1500.00
          4 | Dana Karimova    |   800.00
          5 | Emil Joldoshev   | 12000.00
          6 | Aliya Omarova    |   250.00
          1 | Aigerim Sadykova |  4900.00
          2 | Timur Asanov     |  3100.00
(6 rows)

lab_16=*# COMMIT;
COMMIT
```

READ COMMITTED:

```
lab_16=# BEGIN TRANSACTION ISOLATION LEVEL READ COMMITTED;
BEGIN
lab_16=*# SELECT * FROM accounts WHERE balance > 1000;
 account_id |      owner       | balance
------------+------------------+----------
          3 | Nurlan Bekov     |  1500.00
          5 | Emil Joldoshev   | 12000.00
          1 | Aigerim Sadykova |  4900.00
          2 | Timur Asanov     |  3100.00
(4 rows)

lab_16=*# SELECT * FROM accounts WHERE balance > 1000;
 account_id |      owner       | balance
------------+------------------+----------
          3 | Nurlan Bekov     |  1500.00
          5 | Emil Joldoshev   | 12000.00
          1 | Aigerim Sadykova |  4900.00
          2 | Timur Asanov     |  3100.00
(4 rows)
```

REPEATABLE READ:

```
lab_16=# BEGIN TRANSACTION ISOLATION LEVEL REPEATABLE READ;
BEGIN
lab_16=*# SELECT * FROM accounts WHERE balance > 1000;
 account_id |      owner       | balance
------------+------------------+----------
          3 | Nurlan Bekov     |  1500.00
          5 | Emil Joldoshev   | 12000.00
          1 | Aigerim Sadykova |  4900.00
          2 | Timur Asanov     |  3100.00
(4 rows)

lab_16=*# SELECT * FROM accounts WHERE balance > 1000;
 account_id |      owner       | balance
------------+------------------+----------
          3 | Nurlan Bekov     |  1500.00
          5 | Emil Joldoshev   | 12000.00
          1 | Aigerim Sadykova |  4900.00
          2 | Timur Asanov     |  3100.00
(4 rows)

lab_16=*# COMMIT;
COMMIT
```

SERIALIZABLE:

```
lab_16=# BEGIN TRANSACTION ISOLATION LEVEL SERIALIZABLE;
BEGIN
lab_16=*# SELECT * FROM accounts;
 account_id |      owner       | balance
------------+------------------+----------
          3 | Nurlan Bekov     |  1500.00
          4 | Dana Karimova    |   800.00
          5 | Emil Joldoshev   | 12000.00
          6 | Aliya Omarova    |   250.00
          1 | Aigerim Sadykova |  4900.00
          2 | Timur Asanov     |  3100.00
(6 rows)

lab_16=*# UPDATE accounts SET balance = balance * 1.05;
UPDATE 6
lab_16=*# COMMIT;
COMMIT
```

Setting session-level isolation:

```
lab_16=# BEGIN;
BEGIN
lab_16=*# SET TRANSACTION ISOLATION LEVEL REPEATABLE READ;
SET
lab_16=*# COMMIT;
COMMIT
```

SAVEPOINTS: creating and using

```
lab_16=# BEGIN;
BEGIN
lab_16=*# INSERT INTO customers (name, email) VALUES ('Alice', 'alice@email.com');
INSERT 0 1
lab_16=*# SAVEPOINT after_customer_insert;
SAVEPOINT
lab_16=*# INSERT INTO orders (customer_id, total) VALUES (1, 500.00);
INSERT 0 1
lab_16=*# ROLLBACK TO SAVEPOINT after_customer_insert;
ROLLBACK
lab_16=*# INSERT INTO orders (customer_id, total) VALUES (1, 300.00);
INSERT 0 1
lab_16=*# COMMIT;
COMMIT
```

SAVEPOINTS: multiple savepoints

```
lab_16=# BEGIN;
BEGIN
lab_16=*# INSERT INTO products (name, price) VALUES ('Laptop', 999.99);
INSERT 0 1
lab_16=*# SAVEPOINT sp1;
SAVEPOINT
lab_16=*# INSERT INTO products (name, price) VALUES ('Mouse', 25.99);
INSERT 0 1
lab_16=*# SAVEPOINT sp2;
SAVEPOINT
lab_16=*# INSERT INTO products (name, price) VALUES ('Invalid Product', -50.00);
INSERT 0 1
lab_16=*# ROLLBACK TO SAVEPOINT sp2;
ROLLBACK
lab_16=*# INSERT INTO products (name, price) VALUES ('Keyboard', 79.99);
INSERT 0 1
lab_16=*# COMMIT;
COMMIT
```

SAVEPOINTS: releasing savepoints

```
lab_16=# BEGIN;
BEGIN
lab_16=*# INSERT INTO logs (message) VALUES ('Starting process');
INSERT 0 1
lab_16=*# SAVEPOINT process_start;
SAVEPOINT
lab_16=*# INSERT INTO logs (message) VALUES ('Process completed');
INSERT 0 1
lab_16=*# RELEASE SAVEPOINT process_start;
RELEASE
lab_16=*# COMMIT;
COMMIT
```

Keep Transactions Short and Focused:

```
lab_16=# BEGIN;
BEGIN
lab_16=*# UPDATE inventory SET stock = stock - 1 WHERE product_id = 101;
UPDATE 1
lab_16=*# INSERT INTO order_items (order_id, product_id, quantity) VALUES (1, 101, 1);
INSERT 0 1
lab_16=*# COMMIT;
COMMIT
```

Handle Errors Properly:

```
lab_16=# BEGIN;
BEGIN
lab_16=*# DO $$
lab_16$*# DECLARE
lab_16$*# current_balance DECIMAL;
lab_16$*# BEGIN
lab_16$*# SELECT balance INTO current_balance FROM accounts WHERE account_id = 1;
lab_16$*# IF current_balance < 100 THEN
lab_16$*# RAISE NOTICE 'Transaction failed: Insufficient funds';
lab_16$*# ELSE
lab_16$*# UPDATE accounts SET balance = balance - 100 WHERE account_id = 1;
lab_16$*# UPDATE accounts SET balance = balance + 100 WHERE account_id = 2;
lab_16$*# END IF;
lab_16$*# END $$;
DO
lab_16=*# COMMIT;
COMMIT
```

Appropriate Isolation Levels:

```
lab_16=# BEGIN TRANSACTION ISOLATION LEVEL SERIALIZABLE;
BEGIN
lab_16=*# COMMIT;
COMMIT
lab_16=# BEGIN TRANSACTION ISOLATION LEVEL READ COMMITTED;
BEGIN
lab_16=*# COMMIT;
COMMIT
```

SAVEPOINTS for COMPLEX OPERATIONS:

```
lab_16=# BEGIN;
BEGIN
lab_16=*# INSERT INTO orders (customer_id, total) VALUES (1, 1000.00);
INSERT 0 1
lab_16=*# SAVEPOINT before_items;
SAVEPOINT
lab_16=*# INSERT INTO order_items (order_id, product_id, quantity)
lab_16-*# VALUES (currval('orders_id_seq'), 101, 2);
INSERT 0 1
lab_16=*# INSERT INTO order_items (order_id, product_id, quantity)
lab_16-*# VALUES (currval('orders_id_seq'), 102, 1);
INSERT 0 1
lab_16=*# COMMIT;
COMMIT
```

Monitor Transaction Locks and Deadlocks:

```
lab_16=# SELECT
lab_16-# blocked_locks.pid AS blocked_pid,
lab_16-# blocked_activity.usename AS blocked_user,
lab_16-# blocking_locks.pid AS blocking_pid,
lab_16-# blocking_activity.usename AS blocking_user,
lab_16-# blocked_activity.query AS blocked_statement
lab_16-# FROM pg_catalog.pg_locks blocked_locks
lab_16-# JOIN pg_catalog.pg_stat_activity blocked_activity
lab_16-# ON blocked_activity.pid = blocked_locks.pid
lab_16-# JOIN pg_catalog.pg_locks blocking_locks
lab_16-# ON blocking_locks.locktype = blocked_locks.locktype
lab_16-# AND blocking_locks.database IS NOT DISTINCT FROM blocked_locks.database
lab_16-# AND blocking_locks.relation IS NOT DISTINCT FROM blocked_locks.relation
lab_16-# AND blocking_locks.page IS NOT DISTINCT FROM blocked_locks.page
lab_16-# AND blocking_locks.tuple IS NOT DISTINCT FROM blocked_locks.tuple
lab_16-# AND blocking_locks.virtualxid IS NOT DISTINCT FROM blocked_locks.virtualxid
lab_16-# AND blocking_locks.transactionid IS NOT DISTINCT FROM blocked_locks.transactionid
lab_16-# AND blocking_locks.classid IS NOT DISTINCT FROM blocked_locks.classid
lab_16-# AND blocking_locks.objid IS NOT DISTINCT FROM blocked_locks.objid
lab_16-# AND blocking_locks.objsubid IS NOT DISTINCT FROM blocked_locks.objsubid
lab_16-# AND blocking_locks.pid != blocked_locks.pid
lab_16-# JOIN pg_catalog.pg_stat_activity blocking_activity
lab_16-# ON blocking_activity.pid = blocking_locks.pid
lab_16-# WHERE NOT blocked_locks.granted;
 blocked_pid | blocked_user | blocking_pid | blocking_user | blocked_statement
-------------+--------------+--------------+---------------+-------------------
(0 rows)
```

Explicit Transactions for Multiple Operations:

```
lab_16=# BEGIN;
BEGIN
lab_16=*# INSERT INTO audit_log (action, timestamp) VALUES ('user_creation', NOW());
INSERT 0 1
lab_16=*# INSERT INTO users (name, email) VALUES ('John Doe', 'john@example.com');
INSERT 0 1
lab_16=*# INSERT INTO user_preferences (user_id, theme) VALUES
lab_16-*# (currval('users_id_seq'), 'dark');
INSERT 0 1
lab_16=*# COMMIT;
COMMIT
lab_16=# INSERT INTO audit_log (action, timestamp) VALUES ('user_creation', NOW());
INSERT 0 1
lab_16=# INSERT INTO users (name, email) VALUES ('John Doe', 'john@example.com');
INSERT 0 1
```